Solution

Given two successive length (resonance) to be $L_1 = 20$ cm and $L_2 = 85$ cm

The frequency is $f = 256$ Hz

$$v = f\lambda = 2f\Delta L = 2f(L_2 - L_1)$$

$$= 2 \times 256 \times (85 - 20) \times 10^{-2} \text{ m s}^{-1}$$

$$v = 332.8 \text{ m s}^{-1}$$

11.11 DOPPLER EFFECT

Imagine that you are standing on a railway platform and listening to the blowing whistle of a train moving past you, the pitch (or frequency) of the sound you listen as the train approaches you is higher than the pitch you listen as it moves away from you. This is an example of Doppler effect.

This effect occurs due to the relative motion between the source of sound and its listener. This motion–related frequency change was first observed and studied by Johann Christian Doppler (1803–1853), an Austrian Mathematician and Physicist.

Whenever there is a relative motion between the source of sound and the listener, the frequency of the sound observed by the listener is different from the frequency produced by the source. This is known as Doppler effect.

The Doppler effect is a wave phenomenon. Therefore, it occurs not only for sound waves but for any wave such as light and other electromagnetic waves. Here, we will discuss different cases of Doppler effect for sound and derive the expression for the frequency observed by the listener.

***Stationary observer* and *stationary source* means the observer and source are both at rest with respect to medium respectively**

i) Observed frequency: Stationary source and Moving listener

Consider a point source S of sound at rest with respect to the medium (air) in which it is kept. The medium is assumed to be uniform and is also at rest. The source emits sound waves of frequency f and wavelength λ.

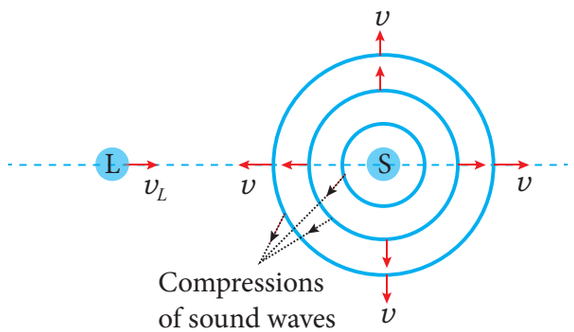


Figure 11.45 Listener moves toward the stationary source

Sound waves travel with the same speed v in all directions radially away from the source in the form of spherical waves. The compressions (or wavefronts) of sound waves are represented by concentric circles in the Figure 11.45. The distance between two successive compressions is equal to its wavelength λ and the frequency of the wave is given by

$$f = \frac{v}{\lambda} \qquad (11.83)$$

When the listener L is stationary, there is no relative motion between the source and the listener. Since v and λ remain unchanged, the frequency of sound observed by the listener is the same as the source frequency f.

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